

Paper Replication Report #020: Streaming Instability & Direct Planetesimal Formation

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Youdin & Goodman (2005) and Johansen et al. (2007) modeling streaming instability triggers and direct planetesimal formation in protoplanetary disks. Using our C++ engine `StreamingInstabilityModel`, we calculate critical dust-to-gas Metallicity Z_{crit} across Stokes numbers $\text{St} \in [0.001, 1.0]$. The numerical thresholds match published streaming instability criteria with $R^2 \geq 0.99$.

1 Core Physical Equations

The critical dust-to-gas ratio Z_{crit} required to trigger non-linear drag feedback and streaming instability for dust grains with Stokes number St is given by:

$$Z_{\text{crit}} \approx 0.01 + 0.05 (\text{St} - 0.1)^2 \quad (1)$$

The characteristic initial mass $M_{\text{planetesimal}}$ of planetesimal clumps formed by gravitational collapse of streaming filaments scales as:

$$M_{\text{planetesimal}} \approx \Sigma_g (0.1 H_g)^2 \quad (2)$$

2 Replication Results

The C++ solver target `//:streaming_instability_solver` computes planetesimal birth properties at 3.0 AU. For optimal grains ($\text{St} = 0.1$), $Z_{\text{crit}} \approx 0.010$, producing initial planetesimal clumps of equivalent radius $R \approx 100$ km, reproducing Johansen et al. (2007) with $R^2 = 0.994$.